

SUPPORT TICKET MANAGEMENT SYSTEM

Software Requirements Specification

Helpdesk / Customer Support Ticketing

Document Version	1.0 (Draft for Review)
Document Status	In Review
Classification	Internal Use Only
Based On	ITIL 4, ITSM, Zendesk & Freshdesk Best Practices

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1. Introduction & Document Purpose

This document defines the complete software requirements for the Support Ticket Management System (STMS), hereafter referred to as the Helpdesk System. It is intended to serve as the authoritative reference for design, development, testing, and deployment of the system.

The system is modelled after industry-leading platforms including Zendesk, Freshdesk, and Jira Service Management, and follows ITIL 4 principles and IT Service Management (ITSM) standards.

1.1 Scope

The Helpdesk System covers:

- End-to-end lifecycle management of every support interaction — from ticket creation to resolution and closure.
- Multi-channel intake: email, web form, phone (manual log), WhatsApp (VIP clients).
- Two-tier resolution model: CSR (Tier 1) and Development/Engineering (Tier 2).
- Automated client communications at every key milestone.
- CSAT (Customer Satisfaction) feedback collection on all closed tickets.
- Integration with the internal Project Management / Development sprint system.
- Reporting and analytics for operational and management visibility.

1.2 Goals & Objectives

#	Objective	Success Metric
G-01	Single source of truth for all support interactions	Zero duplicate tickets; 100% of client issues logged in STMS
G-02	CSR self-resolution rate	80–90% of tickets resolved at Tier 1 without escalation
G-03	Client communication transparency	Automated email sent at every status change within 2 minutes
G-04	SLA compliance	95%+ of tickets handled within agreed response times
G-05	Client satisfaction measurement	CSAT score collected on 100% of closed tickets
G-06	Audit & traceability	Full immutable audit log on every ticket (who, what, when)

1.3 Design Principles

- One Ticket = One Issue: Every distinct client problem has exactly one ticket. No duplicates, no merging of unrelated issues.
- Email-First Communication: All client-facing communication is email-threaded. Clients never need to log into a portal (unless they choose to).

- **Clear Visibility Separation:** Customer-facing comments vs. internal-only notes are strictly separated. Clients never see internal discussions.
 - **Progressive Escalation:** Tickets escalate only when Tier 1 (CSR) cannot resolve. Escalation carries full history.
 - **Automation by Default:** Routine notifications, SLA timers, and reminders are automated. Manual intervention is reserved for decisions.
 - **Immutable Audit Trail:** Every action, status change, comment, and email is logged with actor and timestamp. Nothing is deleted.
 - **Mobile-Accessible:** All interfaces are responsive and usable on mobile devices.
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2. User Roles & Permissions

The system defines five distinct roles with strictly scoped access controls. Role-based access ensures that each user sees and acts upon only the information relevant to their function.

Role	Primary Function	Key Permissions
Client / End-User	Submits and tracks their own issues	View own tickets only; reply via email; submit CSAT feedback; optional: access to self-service portal
CSR (Tier 1)	First point of contact; resolves majority of tickets	Create tickets (own or on behalf of client); view & update assigned tickets; add client-facing replies and internal notes; escalate to Tier 2; resolve and close tickets
Bridge Person (Tier 2 Coordinator)	Liaison between CSR and Development team	Review all escalated tickets and full history; add internal notes; assign to Developer or QA; communicate with client on complex/sprint-deferred tickets; set sprint schedule
Developer / QA Engineer	Implements technical fixes for escalated tickets	View assigned tickets with full history; update ticket status (In Development, In QA, Deployed); add internal notes and resolution summaries; link to development tasks
Admin / Manager	Oversight, reporting, system configuration	Full access to all tickets, reports, SLA configuration, user management, role assignment, force-close any ticket, view all audit logs

2.1 CSR — Raise on Behalf of Client

A core workflow requirement is that CSRs can raise tickets on behalf of clients. This is the most common ticket creation scenario (e.g., client calls in or walks in). The CSR:

- Selects or creates the client profile.
 - Fills in the ticket details from the client interaction.
 - Chooses the channel (Phone, Walk-in, WhatsApp, etc.).
 - Submits the ticket — the system automatically notifies the client via email with the Ticket ID.
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3. Ticket Creation

3.1 Intake Channels

Channel	How It Enters the System	Notes
Email	Client emails support@company.com — system auto-creates ticket and confirms	Thread automatically maintained via Ticket ID in subject line
Web Form	Client fills out form on company website or portal	Fields mapped directly to ticket fields; confirmation email sent immediately
Phone	CSR creates ticket manually during or after call	Call summary logged as internal note; recording attached if permitted
WhatsApp (VIP)	CSR logs conversation and creates ticket; or system integration if configured	Reserved for clients with VIP flag; one-click template messages available
Walk-in	CSR creates ticket on behalf of client	Channel = Walk-in; client still receives automated email confirmation

3.2 Ticket Creation Form — Required Fields

All fields marked Required must be completed before a ticket can be submitted. Optional fields improve routing and reporting but do not block submission.

Field	Required ?	Description & Rules
Title / Subject	Required	Short, clear one-line description of the issue. Max 150 characters.
Description	Required	Detailed rich-text description. Supports formatting, bullet points, code blocks.
Client Name	Required	Linked to the Client/Company database. Searchable by name, company, or email.
Client Company	Required	Auto-filled once Client Name is selected.
Contact Email	Required	Primary address for all automated notifications and email threading.
Intake Channel	Required	Email / Web Form / Phone / WhatsApp / Walk-in. Drives communication defaults.
Category	Required	Bug Report / Feature Request / Service Request / Complaint / General Inquiry / Other.
Priority	Required	Critical / High / Medium / Low. Can be auto-suggested by Category; overridable by CSR.

Field	Required ?	Description & Rules
Assigned CSR	Auto / Manual	Auto-assigned via round-robin or skill-based routing. CSR can re-assign.
Attachments	Optional	Screenshots, log files, videos, documents. Max 20 MB per file.
Internal Notes	Optional	Private notes visible only to staff. Not sent to client.
Raised By	Auto	System records whether raised by client or by CSR on behalf of client.

3.3 Automatic Actions on Ticket Creation

- Unique Ticket ID generated in format: SUP-YYYY-MMDD-NNN (e.g., SUP-2026-0423-001).
- Ticket status set to New and placed in CSR queue immediately.
- Automated confirmation email sent to client within 60 seconds:
 - Subject: [Ticket #SUP-2026-0423-001] Your Support Request Has Been Received
 - Body includes: Ticket ID, title, expected response time based on priority/SLA, support contact.
- Timestamp and intake channel recorded.
- SLA timer starts automatically based on priority level.
- CSR receives in-app and email notification of new ticket assignment.

3.4 Ticket ID Conventions

Format	SUP-{YEAR}-{MMDD}-{SEQ} Example: SUP-2026-0423-001
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The Ticket ID is embedded in all outbound email subject lines. Email replies from clients are automatically threaded to the correct ticket using this ID. No manual linking is required.

4. Ticket Lifecycle & Status Workflow

4.1 Status Definitions

Every ticket has a single active status at all times. The following statuses define the complete lifecycle:

Status	Owner / Tier	Description
New	System / CSR Queue	Ticket has just been created. Awaiting CSR pickup. SLA timer running.
Open	CSR (Tier 1)	CSR has taken ownership and is actively working on the ticket.
Waiting for Client	CSR (Tier 1)	CSR has requested more information from client. SLA clock paused.
Resolved	CSR (Tier 1)	CSR has provided a solution. Client notified. Awaiting confirmation or auto-close.
Escalated to Dev	Bridge Person	CSR could not resolve. Full history forwarded to Bridge Person for review.
Under Review	Bridge Person	Bridge Person is reviewing ticket history and communicating with client/CSR.
Deferred to Sprint	Bridge Person	Issue requires significant development effort; scheduled for a future sprint.
In Development	Developer	Developer has been assigned and is actively working on the fix.
In QA / Testing	QA Engineer	Fix is complete and undergoing quality assurance testing.
Ready for Deployment	Developer / DevOps	QA passed; fix is queued and approved for deployment.
Deployed	System / Developer	Fix has been deployed to production. Client and CSR automatically notified.
Closed	System / CSR / Admin	Ticket fully resolved and closed. CSAT collected or auto-close triggered.
Reopened	Client / CSR	Client indicates issue persists after resolution. Previous history retained.

4.2 Tier 1 Resolution Flow (CSR)

This is the primary and most common path. 80–90% of tickets should be resolved here.

Step	Action	System Behaviour
1	Ticket Created	Status = New. Automated confirmation email sent to client. SLA timer starts.
2	CSR Opens Ticket	Status = Open. CSR reviews full ticket details, history, and any attachments.
3	CSR Investigates	CSR checks Knowledge Base, prior similar tickets, and available resources.
4a	Resolution Found	CSR adds client-facing reply with solution details. Status = Resolved.
4b	More Info Needed	CSR sends client-facing question. Status = Waiting for Client. SLA paused.
5	Client Notified	Automated email: 'Your ticket #XXXX has been resolved. Please confirm or rate.'
6	Feedback Requested	CSAT rating request embedded in email. Client clicks link — no login required.
7	Auto-Close	If no client response within 3 business days, ticket auto-closes with notification.

4.3 Tier 2 Escalation Flow (Development)

Used when the CSR determines the issue is beyond Tier 1 capability: technical bugs, code defects, infrastructure issues, or complex feature requests.

Step	Action	System Behaviour
1	CSR Escalates	CSR clicks 'Escalate to Development'. Status = Escalated to Dev. Bridge Person notified.
2	Bridge Person Reviews	Bridge Person reads full ticket history — all emails, CSR notes, attachments.
3a	Quick Clarification	Bridge Person adds internal note or sends client-facing update requesting details.
3b	Sprint Deferral	If large effort: client emailed — 'This will be addressed in Sprint [X], expected [Date].'
4	Developer Assigned	Bridge Person assigns to Developer or creates linked Dev Task. Status = In Development.
5	Development Work	Developer works on fix. Updates status, adds internal progress notes.
6	QA Testing	Fix moved to QA. Status = In QA / Testing. QA Engineer verifies resolution.

Step	Action	System Behaviour
7	Deployment	Approved fix deployed. Status = Deployed. Automated email to client and CSR.
8	Closure & CSAT	CSAT request sent. Ticket auto-closes after 3 business days if no response.

4.4 Status Transition Rules

The following transitions are enforced by the system. Invalid transitions are blocked.

- New can only transition to: Open, or Closed (Admin only).
 - Open can transition to: Waiting for Client, Resolved, Escalated to Dev.
 - Waiting for Client can transition to: Open (when client replies), Closed (Admin only).
 - Resolved can transition to: Closed, Reopened (if client disputes resolution).
 - Escalated to Dev can transition to: Under Review, Deferred to Sprint, In Development.
 - In Development can transition to: In QA / Testing.
 - In QA / Testing can transition to: Ready for Deployment, In Development (if QA fails).
 - Ready for Deployment can transition to: Deployed.
 - Deployed can transition to: Closed, Reopened.
 - Closed can transition to: Reopened only. No other transitions allowed.
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5. Communication Requirements

5.1 Email Communication (Primary Channel)

Email is the default and primary communication channel for all client-facing interactions. All email communication is threaded by Ticket ID.

5.1.1 Email Threading

- All outbound emails include the Ticket ID in the subject line: [Ticket #SUP-2026-0423-001] {Original Subject}
- Client replies to any email in the thread are automatically ingested and attached to the correct ticket.
- The reply-to address is a per-ticket alias (e.g., ticket-001@support.company.com) so no manual tagging is required.
- All received and sent emails are stored permanently on the ticket timeline.

5.1.2 Automated Email Triggers

Trigger Event	Recipient(s)	Email Content
Ticket Created	Client	Confirmation with Ticket ID, summary, SLA timeline, and support contact.
CSR Replies (client-facing)	Client	CSR reply threaded on original email. Status update included if changed.
Status: Waiting for Client	Client	Request for information with specific questions from CSR.
Ticket Resolved	Client	Resolution summary with CSAT rating link. Instructions to reopen if needed.
Escalated to Development	Client (optional)	Brief note: 'Your request is being reviewed by our technical team.'
Deferred to Sprint	Client	Acknowledgement with expected sprint date and estimated delivery timeline.
Ticket Deployed / Fixed	Client + Assigned CSR	Deployment confirmation with fix summary. CSAT request included.
Auto-Close Warning (Day 2)	Client	Reminder: 'Ticket will auto-close in 1 business day if no response received.'
Ticket Auto-Closed	Client	Confirmation of closure. Option to reopen by replying within 7 days.
SLA Breach Warning	Assigned CSR + Manager	Internal alert: 'Ticket #XXXX is approaching SLA breach.'
Ticket Reopened	CSR + Bridge Person	Internal notification with previous resolution context.

5.2 WhatsApp Communication (VIP Clients)

- Available only for clients with the VIP flag enabled on their profile.
- CSR can send one-click template messages with Ticket ID embedded.
- All WhatsApp messages are logged as client-facing comments on the ticket timeline.
- Templates include: Ticket Created, Awaiting Your Response, Ticket Resolved, and Fix Deployed.
- WhatsApp is a supplementary channel; email remains the authoritative thread.

5.3 Phone Call Logging

- Phone calls do not create tickets automatically.
- CSR creates ticket during or immediately after the call.
- CSR logs a call summary as an internal note (not visible to client).
- If call recording is available and permitted, it is attached to the ticket.
- Client receives automated email confirmation after ticket is created via phone channel.

5.4 Internal Notes vs. Client-Facing Comments

Attribute	Internal Note	Client-Facing Comment
Visible to Client?	No — staff only	Yes — sent via email to client
Visible to Staff?	Yes — CSR, Bridge, Dev, QA, Admin	Yes — all staff
Triggers Email?	No	Yes — immediately
Typical Use	Investigation notes, call summaries, inter-team coordination, Bridge Person assignments	Replies to client, status updates, resolution details, request for info
Marked In UI?	Yellow/orange background with lock icon	White/blue background with send icon

6. Service Level Agreements (SLA)

6.1 Default SLA Tiers

SLA timers start automatically on ticket creation and are enforced by the system. Timers pause when ticket status is Waiting for Client.

Priority	First Response SLA	Resolution SLA	Typical Use Case
Critical	1 Business Hour	4 Business Hours	System down, data loss, security breach, complete service failure
High	4 Business Hours	1 Business Day	Major feature broken, significant client workflow impacted
Medium	1 Business Day	3 Business Days	Partial functionality impacted, workaround available
Low	2 Business Days	7 Business Days	Minor issue, cosmetic, general inquiry, feature request

6.2 SLA Enforcement & Escalation

- At 75% of SLA time elapsed: System sends automated warning email to assigned CSR and their supervisor.
- At 90% of SLA time elapsed: Manager and Bridge Person are notified.
- At 100% (SLA Breach): Ticket is flagged as Breached in the dashboard. Incident logged in SLA compliance report.
- Breached tickets are highlighted in red on all dashboards.
- SLA reports are included in weekly management summaries.

6.3 SLA Pause Conditions

- SLA timer pauses automatically when ticket status is set to Waiting for Client.
 - SLA timer resumes when the client replies or when CSR changes status back to Open.
 - Admin can manually pause/adjust SLA with a mandatory reason logged in the audit trail.
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7. Ticket Closure & CSAT Feedback

7.1 Closure Triggers

- Manual Close by CSR: CSR marks ticket as Resolved; after client confirmation, status moves to Closed.
- Manual Close by Client: Client clicks 'This is resolved' in the feedback email.
- Auto-Close: Ticket automatically closes 3 business days after status = Resolved or Deployed, if no client response.
- Force-Close by Admin: Admin can force-close any ticket with mandatory reason.

7.2 CSAT Survey Design

A Customer Satisfaction (CSAT) survey is automatically triggered on every ticket closure. The survey is embedded in the closure email — no login required.

Rating	1 star = Very Dissatisfied 2 = Dissatisfied 3 = Neutral 4 = Satisfied 5 = Very Satisfied
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- Client can optionally add a free-text comment (max 500 characters).
- The survey link is valid for 14 days after closure.
- Responses are anonymous to the CSR who handled the ticket (visible to managers only, for fairness).
- CSAT score is calculated as: $(4\text{-star} + 5\text{-star responses}) / \text{Total responses} \times 100\%$.

7.3 CSAT Auto-Close Behaviour

Scenario	System Behaviour
Client replies with rating	CSAT recorded. Ticket closes immediately.
Client replies to email without rating	Reply attached to ticket. CSR notified. Ticket remains Resolved until manually closed.
No response within 3 business days	Day 2: Auto-close warning email sent. Day 3: Ticket auto-closes. CSAT marked as 'No Response'.
Client reopens ticket	Status reverts to Reopened. Previous resolution context preserved. New SLA starts.

8. Integration with Development / Project Management System

8.1 Escalation-to-Task Linkage

When a ticket is escalated to Tier 2 and assigned to a Developer, the Bridge Person can create a linked Development Task in the Project Management system. This linkage ensures:

- Full ticket history (all emails, notes, attachments) is accessible from the Development Task.
- Development status updates automatically propagate back to the Support Ticket status.
- When the Developer marks the Task as 'Deployed', the linked ticket status auto-updates to Deployed.
- The ticket remains the client-facing record; the development task is the internal work record.

8.2 Sprint Integration

- Bridge Person can flag a ticket as Deferred to Sprint and associate it with a specific sprint.
- Sprint planning view shows all open support tickets queued for the sprint.
- Release notes generated at sprint end can optionally list resolved Support Ticket IDs.
- Client is automatically notified when their sprint-deferred ticket is included in a sprint.

8.3 Status Synchronisation

Development Task Status	Support Ticket Status (Auto-Update)
Task Created / Assigned	In Development
Moved to QA Column	In QA / Testing
QA Approved / Ready for Deploy	Ready for Deployment
Deployed to Production	Deployed — automated email sent to client and CSR

9. Knowledge Base

9.1 Purpose

The built-in Knowledge Base (KB) is a searchable library of solved issues, FAQs, how-to guides, and workarounds. It enables CSRs to resolve tickets faster by referencing known solutions.

9.2 Functional Requirements

- CSR can search the Knowledge Base from within an open ticket.
 - CSR can insert a KB article link into a client-facing reply.
 - When a ticket is resolved, CSR is prompted to save the solution as a KB article (optional).
 - KB articles have: Title, Category, Tags, Content (rich text), Related Ticket IDs, Author, Last Updated date.
 - Admin can approve, edit, or archive KB articles.
 - Search supports full-text search across title, tags, and content.
 - Most-used and recently-added articles are surfaced on the CSR dashboard.
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10. Reporting & Analytics

10.1 Standard Reports

All reports are auto-generated on a configurable schedule (daily, weekly, sprint-end) and are accessible to Admin and Manager roles.

Report Name	Description & Key Metrics
Ticket Volume Summary	Total tickets created, resolved, and closed per period. Breakdown by category, priority, and channel.
SLA Compliance Report	% tickets resolved within SLA by priority tier. Breach list with root cause notes.
First Contact Resolution Rate	% of tickets resolved by CSR (Tier 1) without escalation. Target: 80–90%.
Average Resolution Time	Mean time from ticket creation to closure, segmented by priority and category.
CSAT Score Report	Average CSAT rating per period, per CSR, per category. Trend graph. Response rate.
Escalation Rate Report	% tickets escalated to Tier 2. Breakdown by CSR, category, and client.
CSR Performance Report	Tickets handled, resolution time, CSAT scores, escalation rate per CSR. For performance reviews.
Open Ticket Ageing Report	All open tickets sorted by age. Highlights tickets approaching or past SLA.
Client Ticket History	Full ticket history per client/company. Useful for account reviews and renewals.
Knowledge Base Usage Report	Most-viewed articles, articles linked to resolved tickets, gap analysis (searches with no results).

10.2 Dashboards

- CSR Dashboard: My open tickets, SLA countdown timers, new assignments, quick-access KB search.
- Manager / Admin Dashboard: Live ticket queue, SLA breach alerts, CSAT trend, team workload heatmap.
- Bridge Person Dashboard: Escalated tickets queue, sprint-linked tickets, Dev team workload.

10.3 Export

- All reports exportable to CSV and PDF.
- Date range and filter options available on all exports.

11. Non-Functional Requirements

11.1 Performance

- Ticket list and dashboard load in under 2 seconds for up to 10,000 active tickets.
- Search returns results in under 1 second.
- Automated emails are sent within 60 seconds of trigger event.

11.2 Availability

- System uptime target: 99.5% (excluding scheduled maintenance).
- Scheduled maintenance windows communicated 48 hours in advance.

11.3 Security & Privacy

- Role-based access control enforced at both UI and API level.
- Clients can only access their own tickets.
- All data transmitted over HTTPS/TLS 1.2+.
- Internal notes are never transmitted to clients under any circumstances.
- Audit logs are immutable and retained for a minimum of 3 years.
- GDPR / applicable data privacy compliance: personal data is handled per the organisation's data policy.

11.4 Usability

- All interfaces are responsive and fully functional on mobile devices (iOS and Android).
- Dark mode supported.
- Accessibility: WCAG 2.1 Level AA compliance target.
- Bulk actions available: bulk assign, bulk close, bulk export.

11.5 Audit & Compliance

- Every action (status change, comment, assignment, email sent/received) is recorded in an immutable audit log.
- Audit log entries include: timestamp, actor (user ID + name + role), action taken, previous state, new state.
- Audit logs accessible to Admin role only.

12. Glossary

Term	Definition
STMS	Support Ticket Management System — the system described in this document.
CSR	Customer Service Representative — Tier 1 support agent.
Bridge Person	The coordinator role between CSR team and Development team. Reviews escalated tickets and manages Tier 2 workflow.
Tier 1	CSR-level support. Handles 80–90% of tickets. First point of resolution.
Tier 2	Development/engineering-level support. Handles escalated technical issues.
CSAT	Customer Satisfaction Score. A 1–5 star rating collected from clients after ticket closure.
SLA	Service Level Agreement. Agreed response and resolution time targets per priority level.
KB	Knowledge Base. Internal library of solved issues and support articles.
ITIL 4	IT Infrastructure Library version 4. A framework of best practices for IT service management.
FCR	First Contact Resolution. % of tickets resolved by CSR without escalation.
Ticket ID	Unique system-generated identifier in format SUP-YYYY-MMDD-NNN.
Internal Note	A comment visible only to staff. Never seen by the client.
VIP Client	A client flagged for premium support, including WhatsApp communication.
Auto-Close	Automatic ticket closure 3 business days after Resolved/Deployed with no client response.
Sprint Deferral	Scheduling a ticket's technical resolution into a future development sprint.

Appendix A: Document Revision History

Version	Date	Author	Changes
1.0	April 2026	—	Initial draft based on stakeholder requirements and best practices review.

— End of Document —